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Accelerating treatments for rare diseases

AI-powered, patient-inspired



The problem

Rare disease patients

7,000 rare diseases

400m patients worldwide

50% are children

Pharma companies rely on blockbuster drugs

R&D costs $2-3bn



Takes 12-14 years



Success rate of 5%

95% of rare diseases don't have an approved treatment

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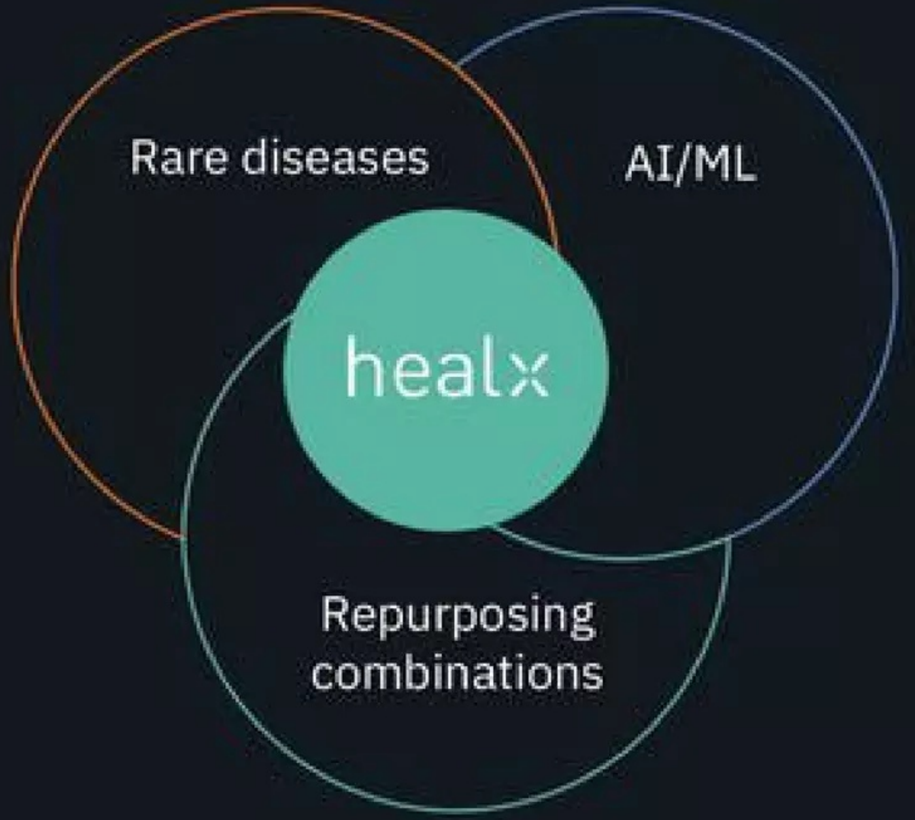
Mission is simple:

100 rare disease treatments towards the clinic by 2025

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AI-powered biotech

Healx repurposes and combines existing drugs using AI to find novel rare disease treatments at scale



Healx redefines and de-risks drug discovery

Traditional drug discovery process



Drug repurposing with Healx



Healx therapeutic pipeline has 10-fold lower attrition rate

Brown, D et al., Drug Discovery Today, 8, (23) 106-1077 (Dec 2003)

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The solution

100 rare disease treatments towards the clinic by 2025

Develop treatments for rare diseases by using repurposing:



AI-powered drug repurposing	Conventional drug discovery
3-5 years	12-14 years
$50m	$2-3bn
25%	5%

Millions of rare disease patients will have an approved treatment for the first time

References: Lin S et al. (2017) Drug Repurposing: The New R&D. Rare Disease Review. Schuhmacher A et al. (2016) Changing R&D Models in Research-Based Pharmaceutical Companies. Journal of Translational Medicine.

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AI-powered drug discovery

Healx knowledge graph

Most comprehensive rare disease knowledge graph:

Clinical, biological and pharmacological information informs and underpins AI treatment discovery.

Integration of public / private databases with proprietary data

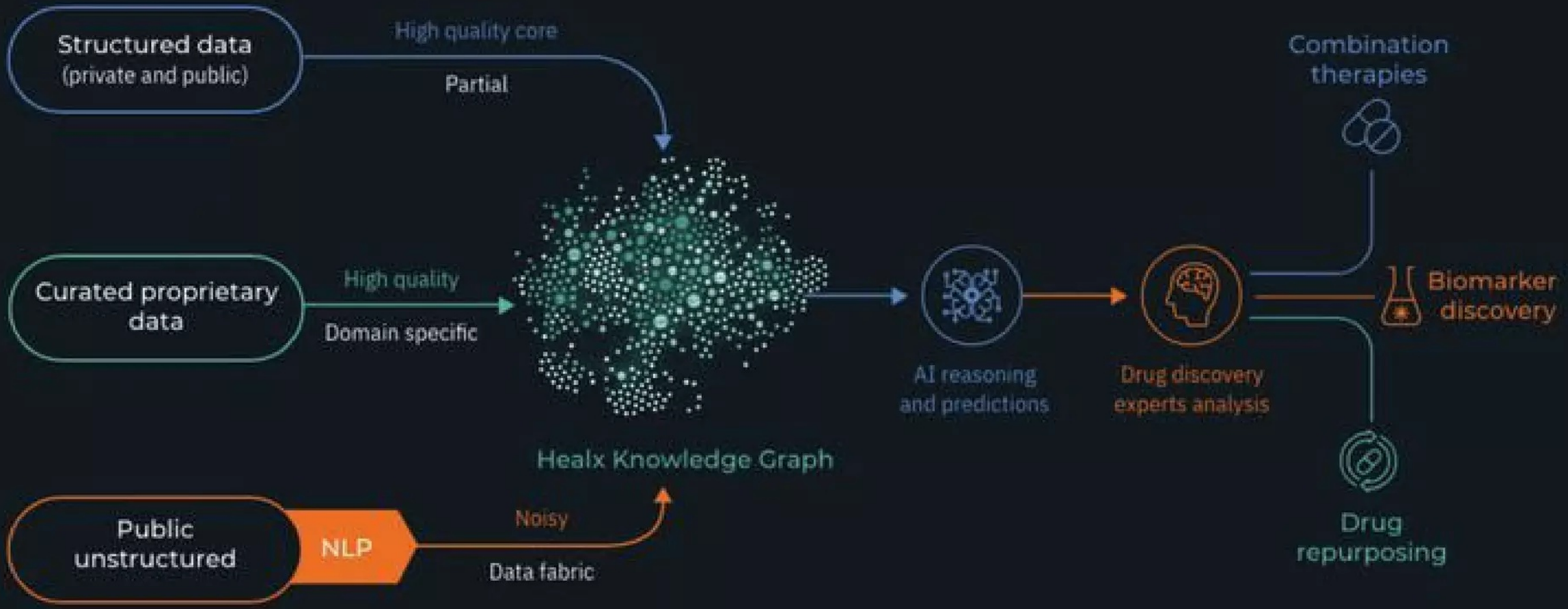
A billion relationships inform treatment discovery

Highly-curated data



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Healnet: from data to drugs



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Therapeutic pipeline

Current projects





The team

Leadership team



Kate Hilyard, PHD

Chief Operating Officer

Senior discovery leadership roles in CRL, Cambridge Antibody Technology and Roche

Over 25 years' drug discovery experience, 10 years' rare disease experience



Tim Guilliams, PHD

Co-founder, Chief Executive

Founding Director, Trustee, Cambridge Rare Disease Network

Cambridge University Graduate

PhD in Biophysics



Neil Thompson, PHD

Chief Scientific Officer

Former SVP of Biology at Astex with over 30 years' drug discovery experience

Progressed more than 10 drugs into patients across a variety of therapeutic areas



Ian Roberts, PHD

Chief Technology Officer

Tech leadership roles in multiple genomics biotech startups

Over 20 years' experience in genomic technologies, strong background in rare cancers



Andrea Pierleoni, PHD

Head of AI

Strong background in NLP, full-stack dev & knowledge graph reasoning

Lead back end architect, Open Targets

PhD in computational biology



Bruce Bloom

Chief Collaboration Officer

Founder, CEO and CSO of non-profit Cures Within Reach

Extensive patient group and charity partnership involvement and over 40 years' healthcare industry experience

Board of directors



Jonathan Milner

Investor Director

Co-founder, Abcam
Co-founder, Milner Institute for Therapeutics



David Brown

Chairman, Founding Director

Ex-Global Head Drug Discovery, Roche

Co-inventor of Viagra



Suranga Chandratillake

Investor Director

Partner, Balderton Capital

Founder and CEO of Blinkx

Former CTO Autonomy



Shaun Grady

Board Director

Global VP, Business Development Operations, AstraZeneca

Led Medimmune acquisition



Irina Haivas

Board Member

Principal at Atomico

Former strategy consultant at Bain & Co. and growth and PE investor at GHO Capital



Tim Guilliams

Founding Director

Founding Director, Trustee, Cambridge Rare Disease Network

Cambridge University Graduate

PhD in Biophysics


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